Day 3: Introduction to IP Addressing

TASK-1 List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Point	IPv4	IPv6
Address size	32 bits	128 bits
Format	Four decimal numbers separated by dots, for example 192.168.1.1	Eight groups of hexadecimal numbers separated by colons, for example 2001:0db8:85a3::8a2e:0370:7334
Number of addresses	About 4.3 billion, which is not enough today	About 3.4 x 10^38, practically unlimited
Header	Variable size, 20 to 60 bytes	Fixed size of 40 bytes, so routers process it faster
Address configuration	Manual or DHCP	Automatic (SLAAC) or DHCPv6, with no need for manual setup
NAT	Widely needed to share one public address	Not needed because every device can have its own public address
Broadcast	Uses broadcast to reach all devices	No broadcast; it uses multicast and anycast
Security	IPsec is optional	IPsec support is built in

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TSAK-2 . Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Network: 192.168.10.0/24, so the subnet mask is 255.255.255.0 and the usable addresses are 192.168.10.1 to 192.168.10.254. The WiFi router is the default gateway, with address 192.168.10.1.

Device	IP address	Subnet mask	Default gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

- Every device has a unique address and all are in the same network, so they can talk to each other.
- The addresses .10 to .13 are kept away from the router's DHCP pool to avoid an address conflict.

Verifying connectivity

- On the laptop: open Command Prompt and run ping 192.168.10.13 (printer) and ping 192.168.10.1 (router). Replies mean that the connection works.
- On Windows, a static IP is set from Settings, Network and Internet, Wi-Fi, Hardware properties, IP assignment, Edit, Manual.

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TASK-3 Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each

Calculation

- 192.168.1.0 is a Class C network with default mask /24. With /27, the borrowed bits = 27 - 24 = 3.
- Number of subnets = 2^3 = 8.
- Host bits = 32 - 27 = 5, so total addresses per subnet = 2^5 = 32 and usable hosts = 32 - 2 = 30.
- Subnet mask = 255.255.255.224 (last octet 11100000 = 128 + 64 + 32 = 224).
- Block size = 256 - 224 = 32, so the subnets start at 0, 32, 64, 96, 128, 160, 192, 224.

Subnet	Network address	Usable host range	Broadcast address	Hosts
1	192.168.1.0/27	192.168.1.1 to 192.168.1.30	192.168.1.31	30
2	192.168.1.32/27	192.168.1.33 to 192.168.1.62	192.168.1.63	30
3	192.168.1.64/27	192.168.1.65 to 192.168.1.94	192.168.1.95	30

Answer: 8 subnets (all 8 are usable in modern networks) and 30 usable hosts per subnet. The first three subnets are shown above.

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TASK-3 You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

Step 1: Convert the mask to CIDR

255.255.255.0 = 11111111.11111111.11111111.00000000, which has 24 ones, so the CIDR notation is /24.

Step 2: Network address (IP AND mask)

	Octet 1	Octet 2	Octet 3	Octet 4
IP address	10	0	5	17 (00010001)
Subnet mask	255	255	255	0 (00000000)
AND result	10	0	5	0 (00000000)

Network address = 10.0.5.0

Step 3: Broadcast address

The host part is the last octet (8 bits). Setting all host bits to 1 gives 11111111 = 255. Broadcast address = 10.0.5.255.

Step 4: Valid host range

First host = network address + 1 = 10.0.5.1. Last host = broadcast address - 1 = 10.0.5.254. Total usable hosts = 2^8 - 2 = 254.

Item	Value
CIDR	/24
Network address	10.0.5.0
Broadcast address	10.0.5.255
Valid host range	10.0.5.1 to 10.0.5.254
Usable hosts	254

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TASK-4 Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Screenshot of my calculation:



Device	Static IPv4 Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Item	My manual answer	Calculator output (expected)
Network address	10.0.5.0	10.0.5.0
Broadcast address	10.0.5.255	10.0.5.255
Usable host range	10.0.5.1 to 10.0.5.254	10.0.5.1 to 10.0.5.254
Usable hosts	254	254